## Supplemental Note 1

### scTrio-seq2 Gene expression data preprocessing

The gene expression data was in two units: FPKM (510 samples, 25,373 genes) and TPM (688 samples, 23,457 genes). We converted all values to TPM to standardize the data. We only kept genes that are common in both units. The final dataset had 1,198 samples across five classes: Primary Tumor (PT, n=616), Lymph Node Metastasis (LN, n=224), Liver Metastasis (ML, n=226), Post-Treatment Liver Metastasis (MP, n=102), and Normal Colon (NC, n=20). We excluded NC from further analysis to mitigate class imbalance due to its small sample size. We performed differential expression (DE) analysis using MAST, a statistical method well-suited for single-cell transcriptomics, to select the important features (genes) for the downstream task. DE analysis was conducted under four conditions: PT vs. Rest, ML vs. Rest, MP vs. Rest, and LN vs. Rest. We selected significant DE genes based on adjusted p-value $< 0.05$ and log2 fold change $> 2$. The number of DE genes identified per comparison were: 214 (PT), 675 (ML), 1,395 (MP), and 503 (LN). We took the union of these sets, resulting in 1,673 unique differentially expressed genes. The volcano plots of DE analysis are shown in Supplemental Figure 1. To assess how these DE genes improve class separation, we visuazlied a t-SNE plot using the entire gene expression data and only DE gene expression data. As expected, we observed a better separation between classes in the t-SNE plot for DE genes (Supplemental Figure 2).

### scTrio-seq2 DNA methylation data preprocessing

DNA methylation data were provided at single base resolution. To calculate DNA methylation at the gene level, we calculated the average methylation signal in the promoter region of genes. Gene annotation data was downloaded from the UCSC Table Browser (genome: Human, assembly: GRCh38/hg38, track: NCBI RefSeq). For each gene, we extended the promoter region 1,000 base pairs upstream of the transcription start site and 500 base pairs downstream of the transcription end site. Using these extended regions, we calculated the average DNA methylation rate by aggregating methylation values across the defined coordinates for each gene in each sample. This process was repeated for 1,295 samples and for 22,910 genes. If a gene appeared multiple times with different average methylation rates, we kept only the gene with the highest average value to ensure consistency. At last, we had 1295 samples and 17,123 genes. We then took the top 10% highly variable genes across all samples.

## Supplemental Note 2

### Dataset Splitting and Model Evaluation Technique

For the scTrio-seq2, TCGA-GBM, and TCGA-COAD datasets, we randomly split the samples into 80% for training and 20% for testing. For the TCGA-BRCA and ROSMAP datasets, as training and test sets were provided in earlier work, we kept the original splits to ensure a more fair comparison. In both cases, model training was conducted on the training set using five-fold cross-validation to select the best hyperparameters. The final model was then evaluated on the test set 10 times, and the mean and standard deviation of the performance metrics were reported. For a fair comparison, we applied the same procedure to both the baseline and state-of-the-art (SOTA) methods.

**Supplemental Note 3**

**Model interpretation using LIME**

LIME[1] is a model-agnostic explainability framework to identify the most important features in prediction. We utilized LIME on models trained on ROSMAP and TCGA-GBM datasets to find most important features. We selected top 50 important features for both gene expression branch and DNA methylation branch. GO enrichment analysis of the top 50 gene expression data features in ROSMAP dataset yielded 33 significant GO terms. Among those, six of them (GO:0005875, GO:0008017, GO:0005871, GO:0003777, GO:0098935, GO:0099519) is related to Alzheimer's disease (AD). Studies show that disruption of cytoskeletal processes such as the microtubule-associated complex (GO:0005875), microtubule binding (GO:0008017), and kinesin complex (GO:0005871) with associated microtubule motor activity (GO:0003777) impairs axonal transport of APP and destabilizes microtubule–tau interactions in AD[2-4]. Likewise, perturbations in dendritic transport (GO:0098935) and dense-core vesicle transport (GO:0099519) contribute to mislocalized RNAs and vesicle trafficking deficits that underlie synaptic dysfunction[5-6].

KEGG enrichment of the top 50 DNA methylation data features in ROSMAP dataset yielded 79 significant KEGG pathways. Among them 12 pathways have relevance to AD. KEGG enrichment analyses have consistently highlighted hsa05010 as a top-significant pathway in AD cohorts and models, supporting its use as a reference set of AD genes and interaction modules[7-10]. In parallel, hsa05022 encompasses overlapping mechanisms of neurodegeneration, including APP and VCP, and has been reported as enriched in AD-related datasets and cerebral organoid models[11-13]. Dysregulation of autophagy (hsa04140), mTOR (hsa04150), and PI3K–Akt (hsa04151) signaling alters Aβ and tau clearance, synaptic plasticity, and neuronal survival[14-16]. Apoptosis (hsa04210), Toll-like receptor (hsa04620), TNF (hsa04668), and cytokine–cytokine receptor interaction (hsa04060) pathways highlight the role of neuroinflammation and programmed cell death in AD brains[17-19]. In parallel, disruptions in energy metabolism such as glycolysis/gluconeogenesis (hsa00010), phenylalanine metabolism (hsa00360), tryptophan metabolism (hsa00380), glutathione metabolism (hsa00480), and peroxisome (hsa04146) contribute to oxidative stress and hypometabolism, both early hallmarks of AD[20-21]. Finally, impairment of synaptic and cytoskeletal signaling including MAPK (hsa04010), Notch signaling pathway (hsa04330), axon guidance (hsa04360), regulation of actin cytoskeleton (hsa04810), and focal adhesion (hsa04530) accelerates dendritic spine loss and neuronal connectivity failure[22-23].

For GBM dataset, GO enrichment of the top 50 gene expression features yielded five significant GO terms, all of which are related to GBM. Nuclear receptor activity (GO:0004879) is relevant because activating PPAR-γ restrains GBM growth and enhances temozolomide efficacy in orthotopic models[24], while ligand-modulated transcription factor activity (GO:0098531) applies via the androgen receptor, which drives radiation resistance and creates a vulnerability to anti-androgens[24]. Calmodulin binding (GO:0005516) emerges in GBM transcriptomic analyses as a significantly enriched molecular function among differentially expressed genes[26]. Nuclear steroid receptor activity (GO:0003707) is supported by estrogen receptor-β acting as a tumor suppressor that sensitizes GBM to chemotherapy[27], and steroid binding (GO:0005496) is pertinent because estradiol promotes GBM cell proliferation, migration and invasion in an *EZH2*-dependent manner[28]

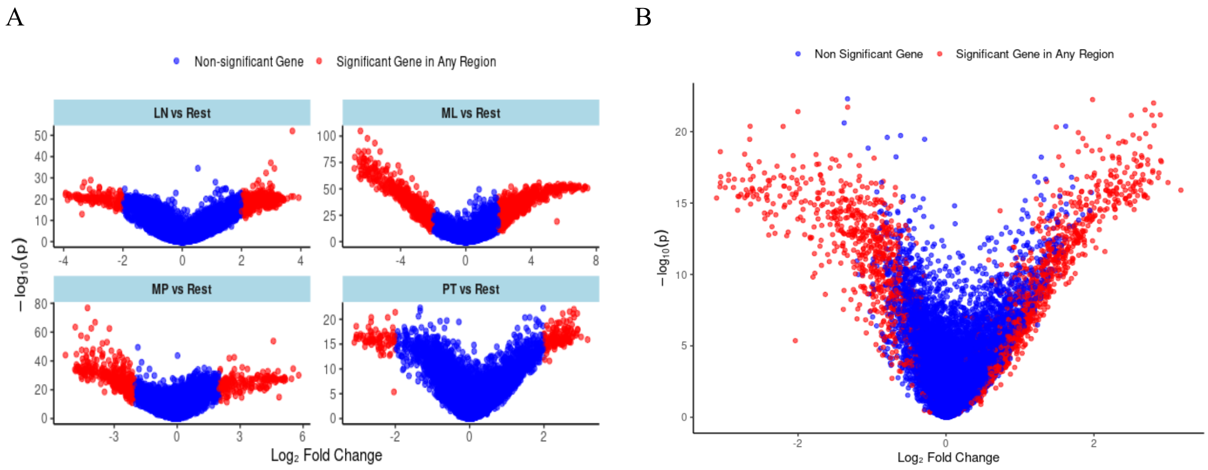
KEGG enrichment of the top 50 DNA methylation features in GBM dataset yielded 16 significant KEGG terms, 15 of which are related to GBM. PI3K–Akt signaling (hsa04151), promoter methylation can silence the tumor suppressor PTEN, a canonical driver of PI3K–Akt activation in GBM[29]. JAK–STAT signaling (hsa04630) is epigenetically tuned via SOCS3: its promoter hypermethylation disables this negative regulator and promotes STAT3-dependent invasion in subsets of gliomas[30]. Cytokine–cytokine receptor interaction (hsa04060) is affected by DNA methylation of innate/adaptive nodes—STING/TMEM173 promoter hypermethylation dampens pathway competence in GBM, and hypermethylation-mediated IL7R regulation has been linked to glioma biology[31-32]. The lysosome/autophagy axis (hsa04142) carries methylation marks on upstream autophagy genes: ULK2 is hypermethylated and silenced in glioma, functionally constraining autophagy-to-lysosome flux[33]. Nucleotide metabolism (hsa01232) and pyrimidine metabolism (hsa00240) tie metabolism to the methylome—IDH3α rewires one-carbon flux, shifting S-adenosylmethionine supply and altering DNA methylation while coordinating nucleotide synthesis in GBM models[34]. Biosynthesis of cofactors (hsa01240) also connects to methyl-group "economy": NNMT creates a "methyl-donor sink," lowering global DNA methylation in GBM stem-like cells[35]. Hormone signaling (hsa04081) shows epigenetic coupling, with androgen-receptor expression associated with genome-wide CpG methylation patterns in glioma/GBM datasets[36]. RNA polymerase (hsa03020) links through MGMT: chromatin-level control of MGMT in GBM includes reduced BRD4/Pol II engagement at a methylated promoter, impacting transcriptional responsiveness to temozolomide[37]. Hematopoietic cell lineage (hsa04640) relevance emerges from GBM-conditioned immune cells, where tumor-infiltrating $CD4^+$ T cells acquire differential DNA methylation programs[38]. Finally, efferocytosis (hsa04148) contributes to the immunosuppressive GBM microenvironment; efferocytosis-derived metabolites can drive DNMT3A-dependent methylation programs in macrophages, and GBM reviews highlight this pathway as a therapeutic target[39-40].

**Supplemental Table 1.** Random Model Comparison in scTrio-seq2 data

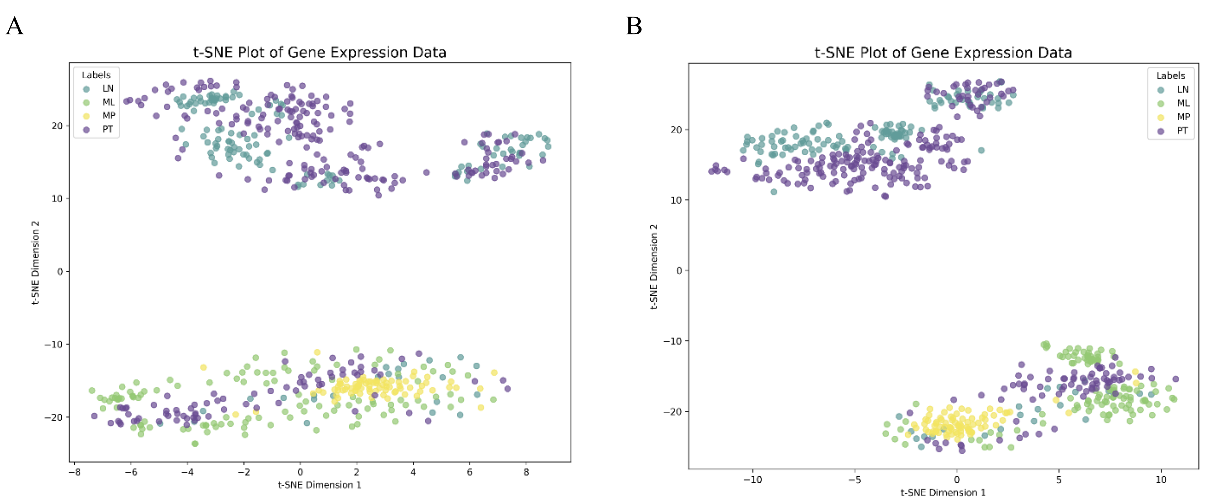
| Model | Precision | Recall | F1 score | Accuracy |
|---|---|---|---|---|
| GE_PDI (Random) | 0.90 ± 0.0310 | 0.88 ± 0.0538 | 0.87 ± 0.0698 | 0.87 ± 0.0590 |
| GE_PDI | 0.93 ± 0.0113 | 0.92 ± 0.0104 | 0.92 ± 0.0105 | 0.92 ± 0.0096 |
| DM_PDI (Random) | 0.90 ± 0.0098 | 0.90 ± 0.0082 | 0.90 ± 0.0072 | 0.92 ± 0.0041 |
| DM_PDI | 0.93 ± 0.0087 | 0.92 ± 0.0098 | 0.92 ± 0.0095 | 0.93 ± 0.0075 |
| GE_PPI (Random) | 0.88 ± 0.0021 | 0.90 ± 0.0014 | 0.88 ± 0.0020 | 0.89 ± 0.0025 |
| GE_PPI | 0.89 ± 0.0658 | 0.89 ± 0.0600 | 0.88 ± 0.0792 | 0.88 ± 0.0647 |
| DM_PPI (Random) | 0.86 ± 0.0112 | 0.85 ± 0.0194 | 0.85 ± 0.0188 | 0.87 ± 0.0111 |
| DM_PPI | 0.91 ± 0.0059 | 0.91 ± 0.0089 | 0.91 ± 0.0079 | 0.92 ± 0.0074 |
| GE_DM_PDI (Random) | 0.92 ± 0.0073 | 0.91 ± 0.0082 | 0.91 ± 0.0087 | 0.91 ± 0.0082 |
| GE_DM_PDI | 0.91 ± 0.0134 | 0.91 ± 0.0096 | 0.90 ± 0.0124 | 0.90 ± 0.0120 |
| GE_DM_PPI (Random) | 0.77 ± 0.1819 | 0.79 ± 0.1586 | 0.76 ± 0.1872 | 0.82 ± 0.1186 |
| GE_DM_PPI | 0.92 ± 0.0240 | 0.91 ± 0.0316 | 0.91 ± 0.0319 | 0.92 ± 0.0267 |
| BioLM-NET (Random PDI, PPI connections) | 0.53 ± 0.1278 | 0.60 ± 0.0688 | 0.53 ± 0.0907 | 0.67 ± 0.0526 |
| BioLM-NET (Random PDI, PPI, pathway connections, LLM embedding) | 0.88 ± 0.0299 | 0.90 ± 0.0287 | 0.88 ± 0.0371 | 0.89 ± 0.0538 |

**Supplemental Table 2.** Significant GO terms from the top 50 SHAP-ranked genes in the gene expression branch in ROSMAP data

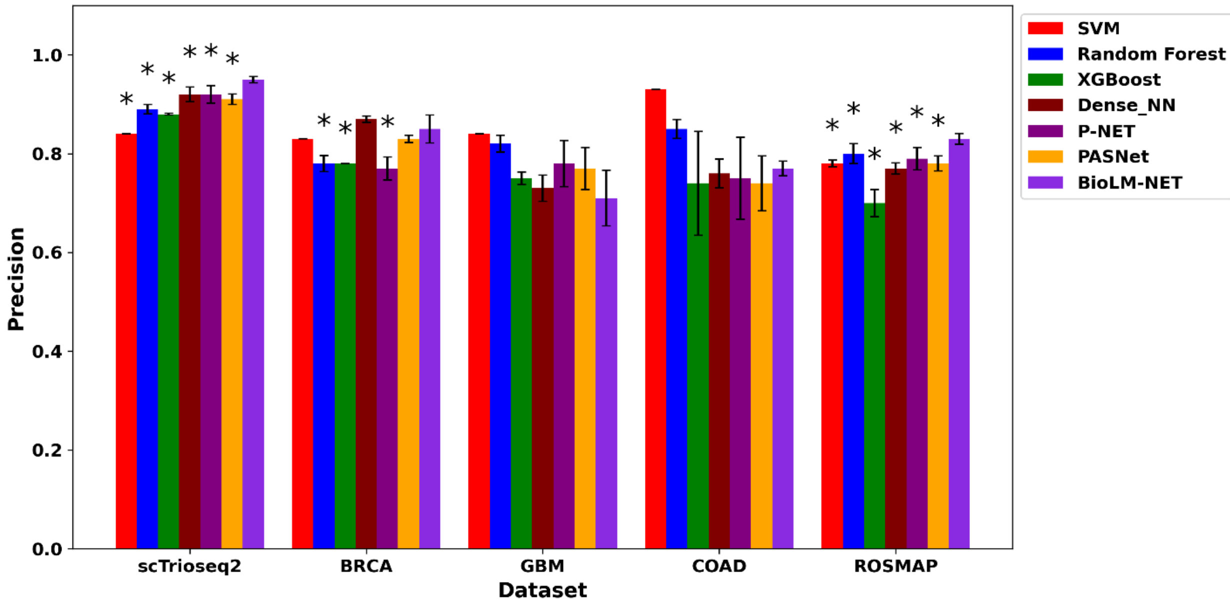
| ID | Description |
|---|---|
| GO:0099519 | dense core granule cytoskeletal transport |
| GO:1901950 | dense core granule transport |
| GO:0008090 | retrograde axonal transport |
| GO:0032253 | dense core granule localization |
| GO:0032252 | secretory granule localization |
| GO:0047496 | vesicle transport along microtubule |
| GO:0008089 | anterograde axonal transport |
| GO:0098937 | anterograde dendritic transport |
| GO:1905383 | protein localization to presynapse |
| GO:0099518 | vesicle cytoskeletal trafficking |
| GO:0098935 | dendritic transport |
| GO:0098840 | protein transport along microtubule |
| GO:0099118 | microtubule-based protein transport |
| GO:0098930 | axonal transport |
| GO:0051145 | smooth muscle cell differentiation |
| GO:0008088 | axo-dendritic transport |
| GO:0032011 | ARF protein signal transduction |
| GO:0032012 | regulation of ARF protein signal transduction |
| GO:0072384 | organelle transport along microtubule |
| GO:0060004 | reflex |
| GO:0005871 | kinesin complex |
| GO:1904115 | axon cytoplasm |
| GO:0035253 | ciliary rootlet |
| GO:0120111 | neuron projection cytoplasm |
| GO:0099524 | postsynaptic cytosol |
| GO:0032839 | dendrite cytoplasm |
| GO:0099522 | cytosolic region |
| GO:0008574 | plus-end-directed microtubule motor activity |
| GO:0008017 | microtubule binding |
| GO:0003777 | microtubule motor activity |
| GO:0015631 | tubulin binding |
| GO:0003774 | cytoskeletal motor activity |



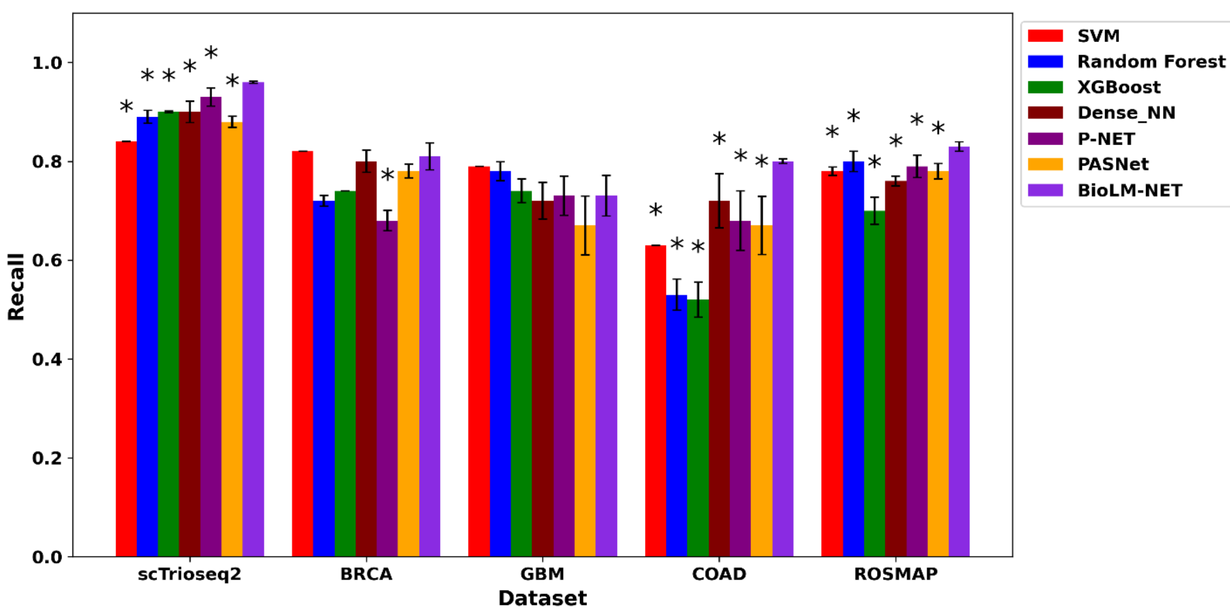

**Supplemental Fig. 1.** Differential Expression (DE) analysis n scTrio-seq2 data. **A.** Volcano plot for DE analysis between LN vs. Rest, ML vs. Rest, MP vs. Rest and PT vs. Rest and **B.** Volcano plot for the final DE genes



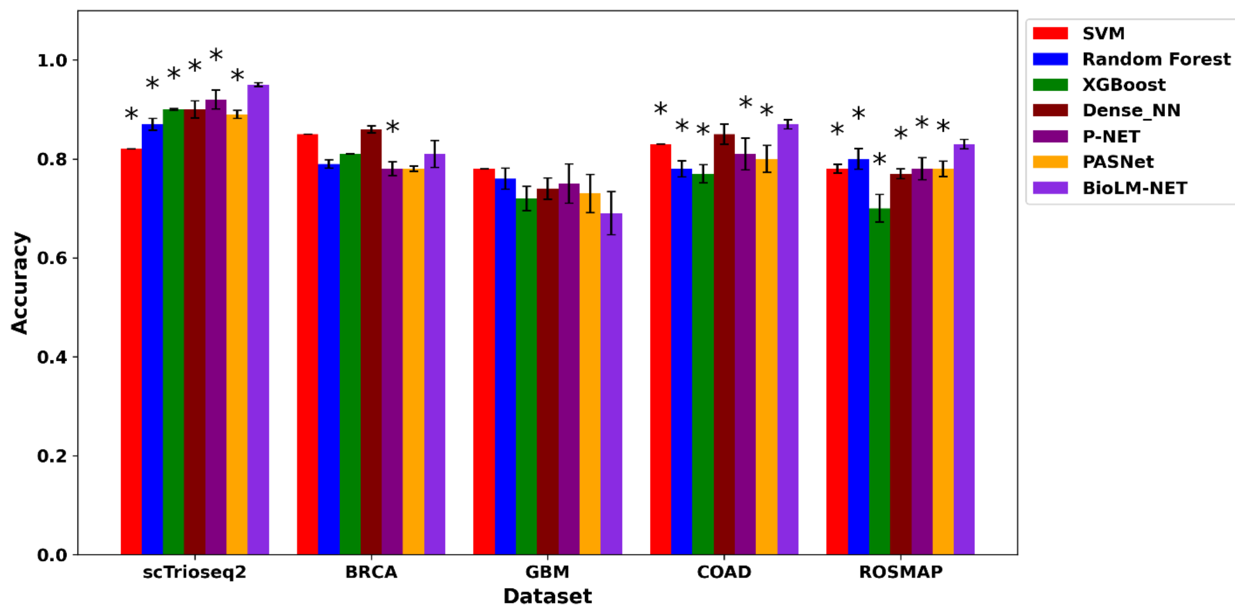

**Supplemental Fig. 2. A.** t-SNE plot for the entire gene expression in scTrio-seq2 data. **B.** t-SNE plot for DE genes only. [PT = Primary Tumor, LN = Lymph Node Metastasis, MP= Post-Treatment Liver Metastasis, ML= Liver Metastasis]



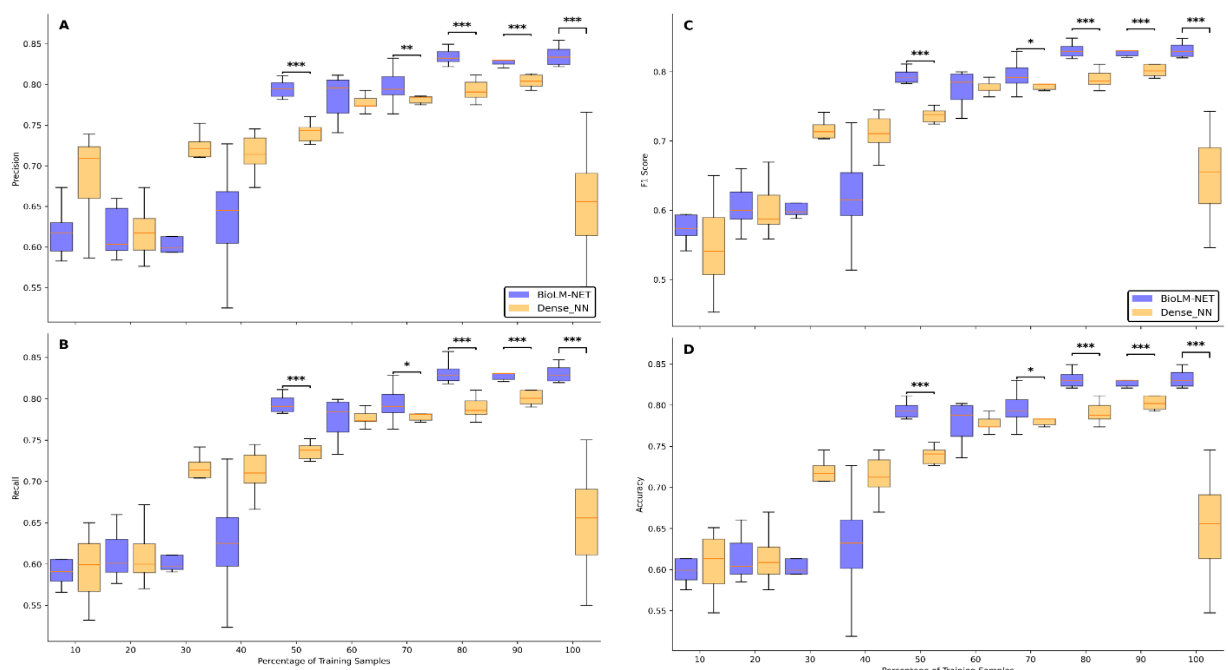

**Supplemental Fig. 3.** Precision comparison of BioLM-NET with baseline and SOTA models. Wilcoxon rank sum test was performed by comparing BioLM-NET with other models (p-value < 0.05 is denoted by *)



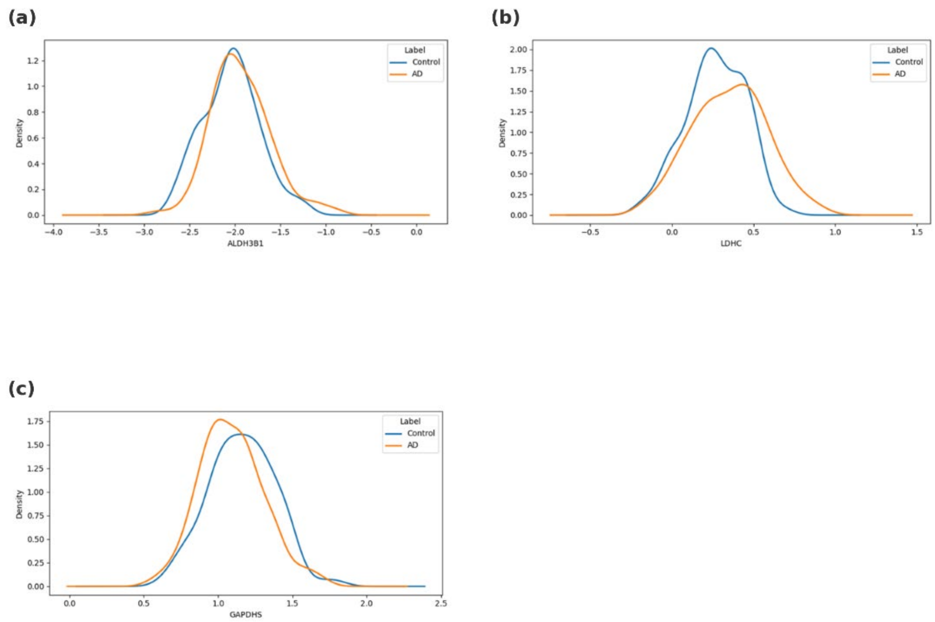

**Supplemental Fig. 4.** Recall comparison of BioLM-NET with baseline and SOTA models. Wilcoxon rank sum test was performed by comparing BioLM-NET with other models (p-value < 0.05 is denoted by *)



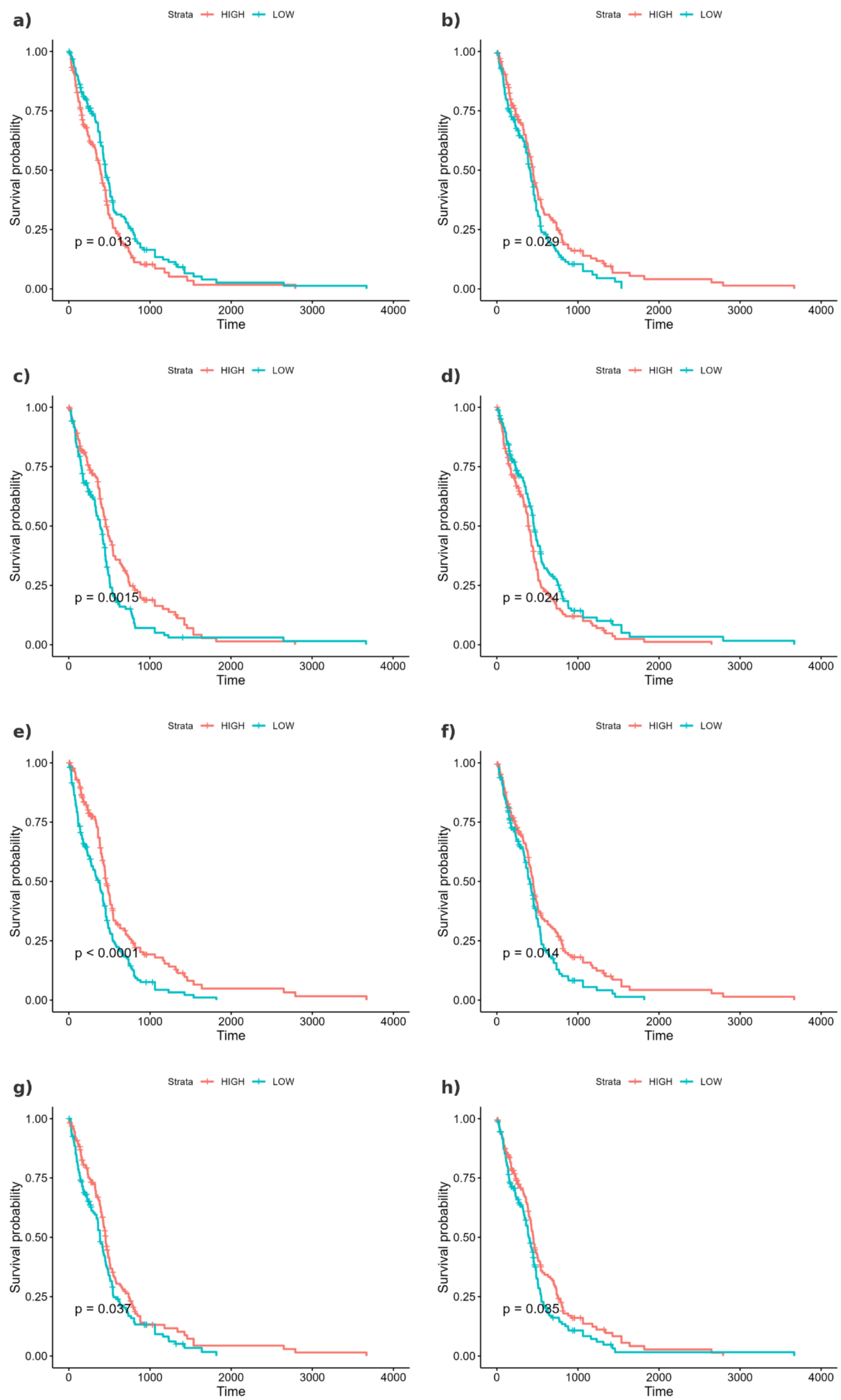

**Supplemental Fig. 5.** Accuracy comparison of BioLM-NET with baseline and SOTA models. Wilcoxon rank sum test was performed by comparing BioLM-NET with other models (p-value < 0.05 is denoted by *)



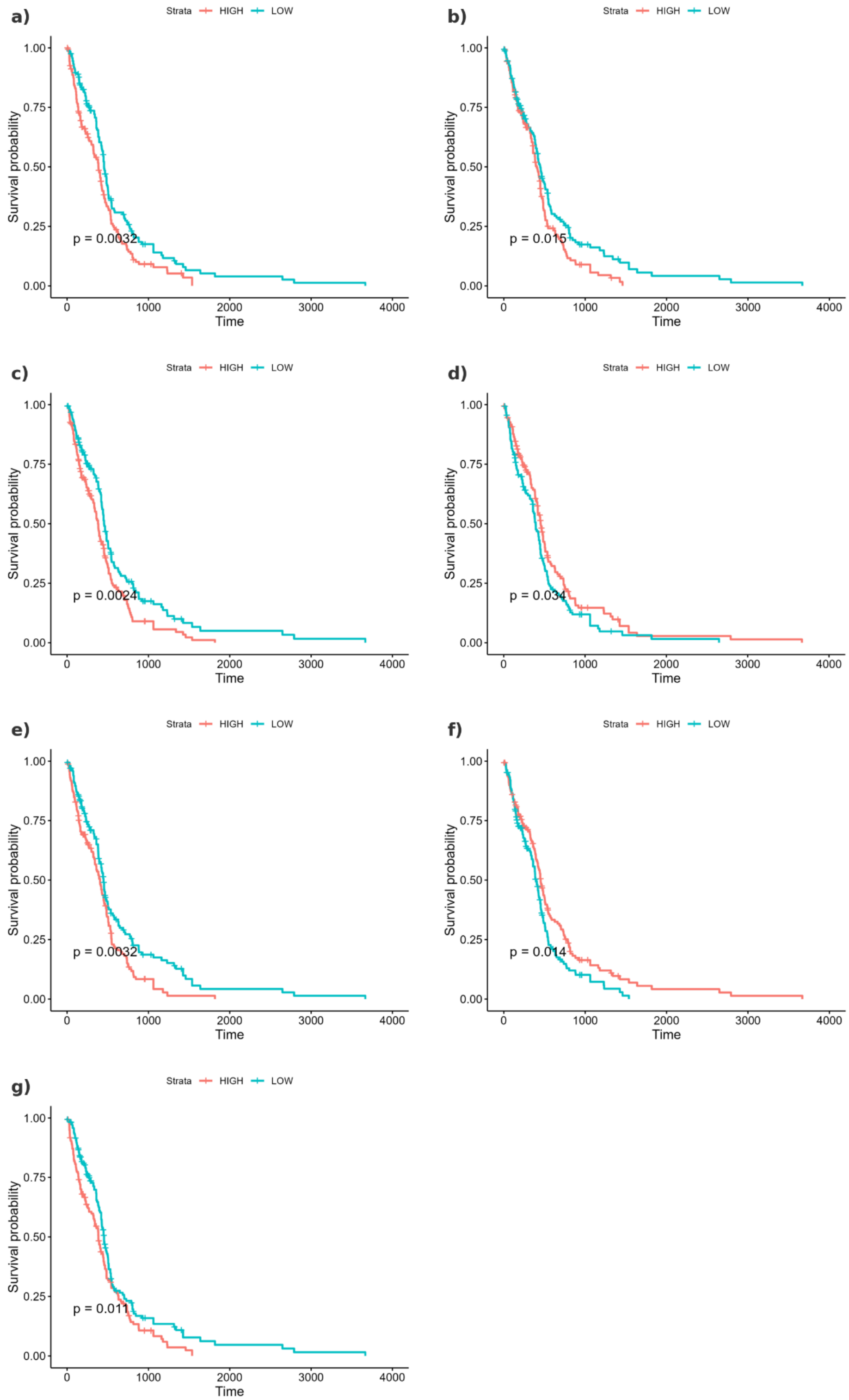

**Supplemental Fig. 6.** Boxplots of (A)Precision (B) Recall (C) F1 score (D) Accuracy for BioLM-NET (blue) and Dense_NN (orange) across 10 independent runs at each percentage of training samples. Statistical significance was assessed with respect to BioLM-NET with an one-sided Wilcoxon rank sum test (***: p-value < 0.001, **: p-value < 0.01, *: p-value < 0.05)



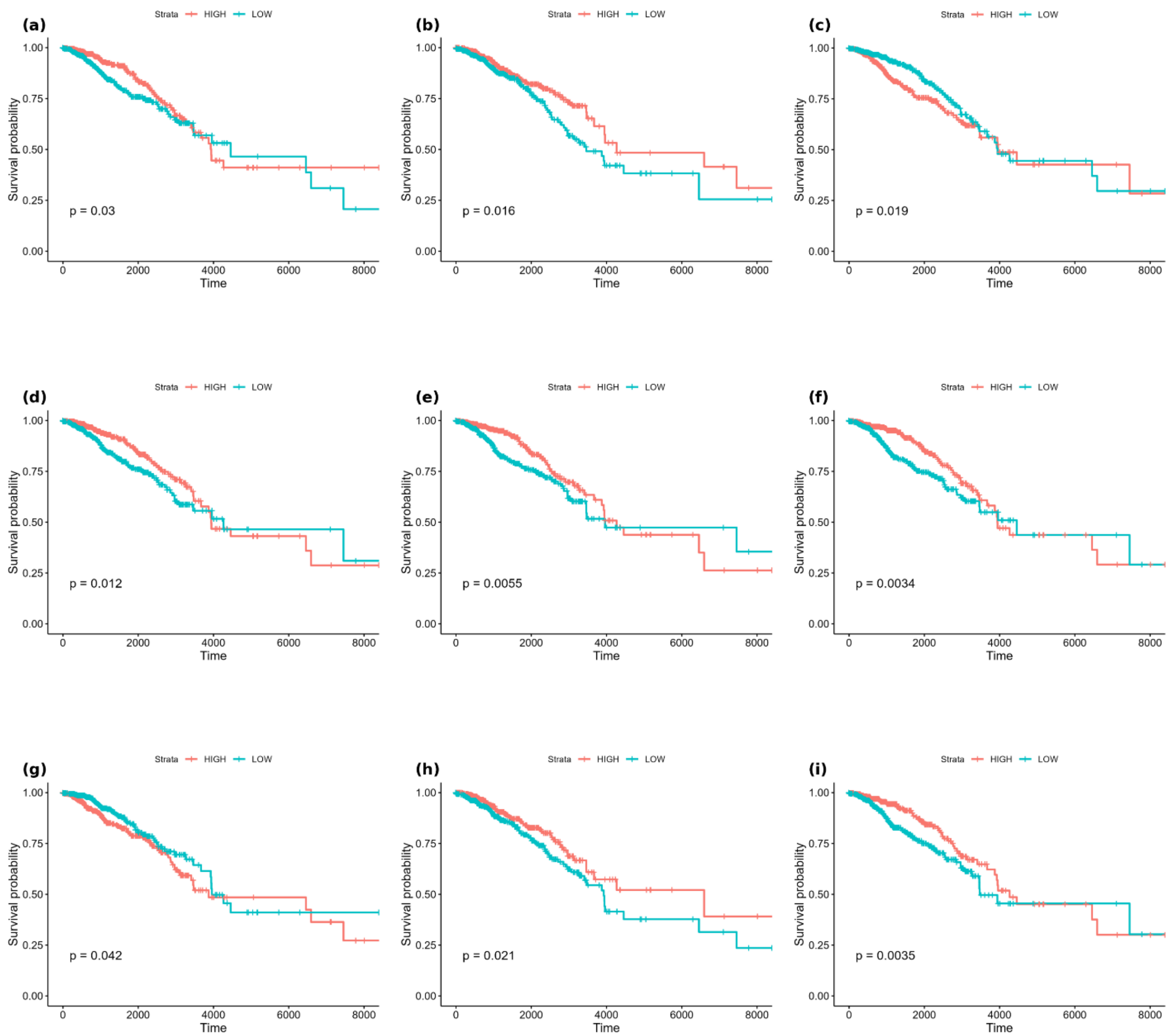

**Supplemental Fig. 7.** DNA methylation values (M-value) between Control vs. AD patients in three genes that are derived from ROSMAP DNA methylation SHAP-ranked top 50 gene.



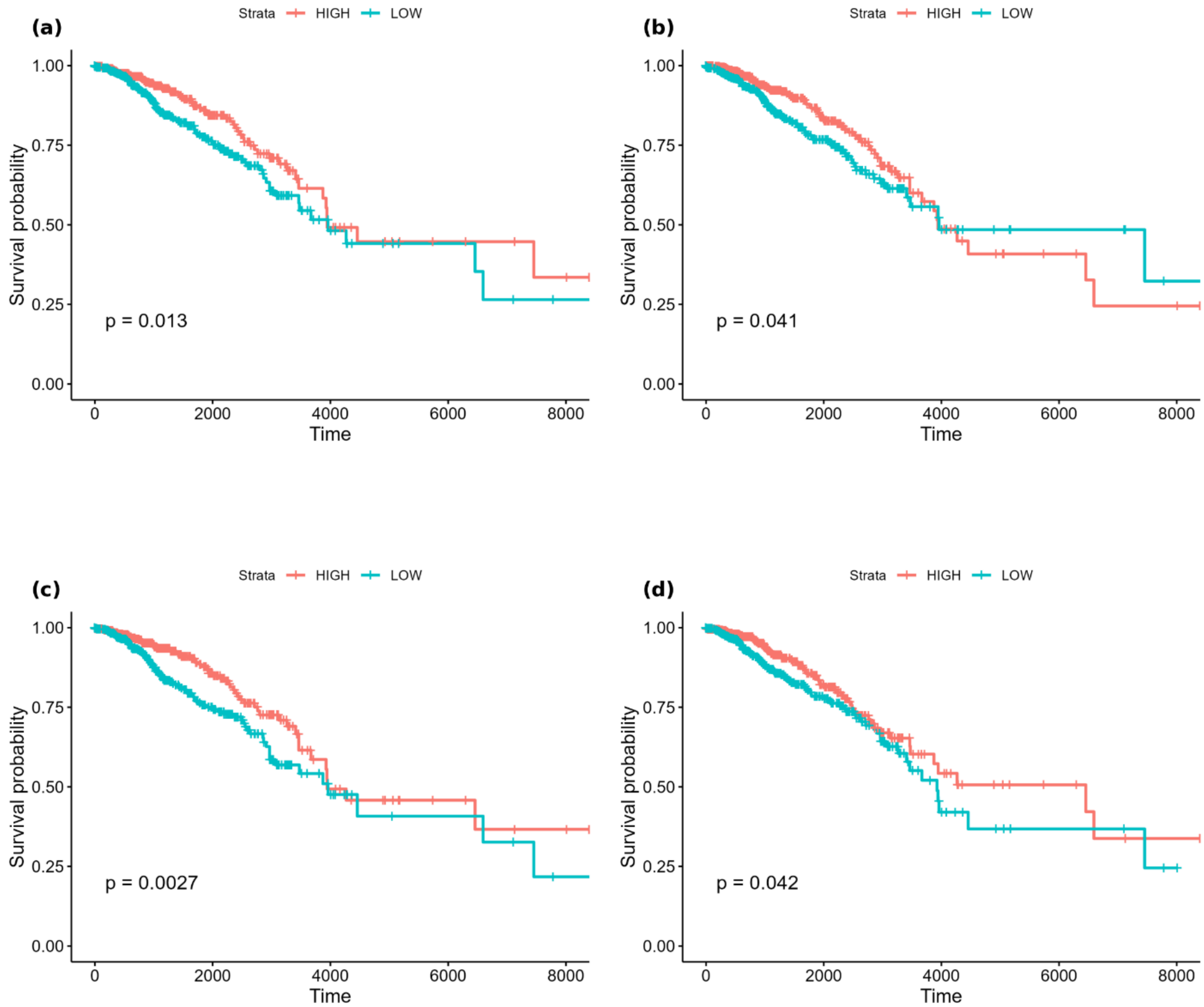

**Supplemental Fig 8.** Kaplan-Meier Plot for GBM (Gene expression features) : Kaplan–Meier survival curves for TCGA-GBM patients stratified by high vs. low expression of a. *CBX7, b. CENPJ, c. DACH1, d. NUDT22, e. PM20D2, f. RALGAPA1P1, g. TBX3, h. WDPCP*.



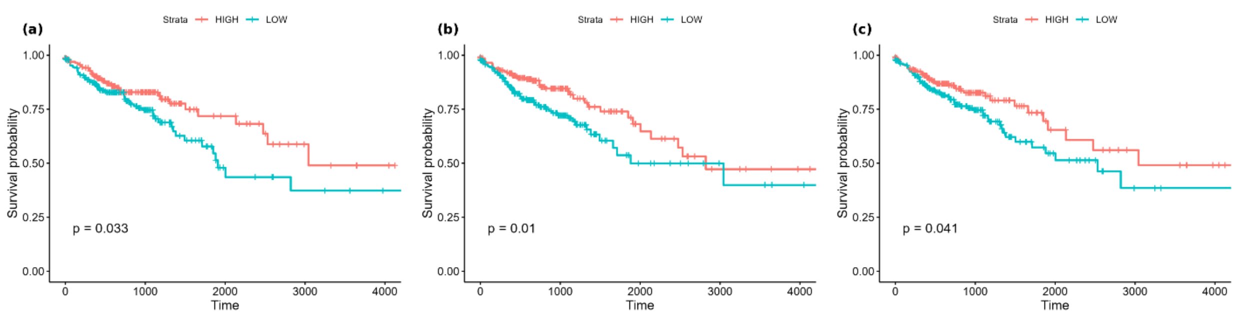

**Supplemental Fig 9.** Kaplan-Meier Plot for GBM (DNA Methylation features) patients stratified by high vs. low expression of (a) *C16orf89*, (b) *CANT1*, (c) *CDKN2D*, (d) *CPS1*, (e) *ELSPBP1*, (f) *GON4L*, and (g) *LDLRAP1*.



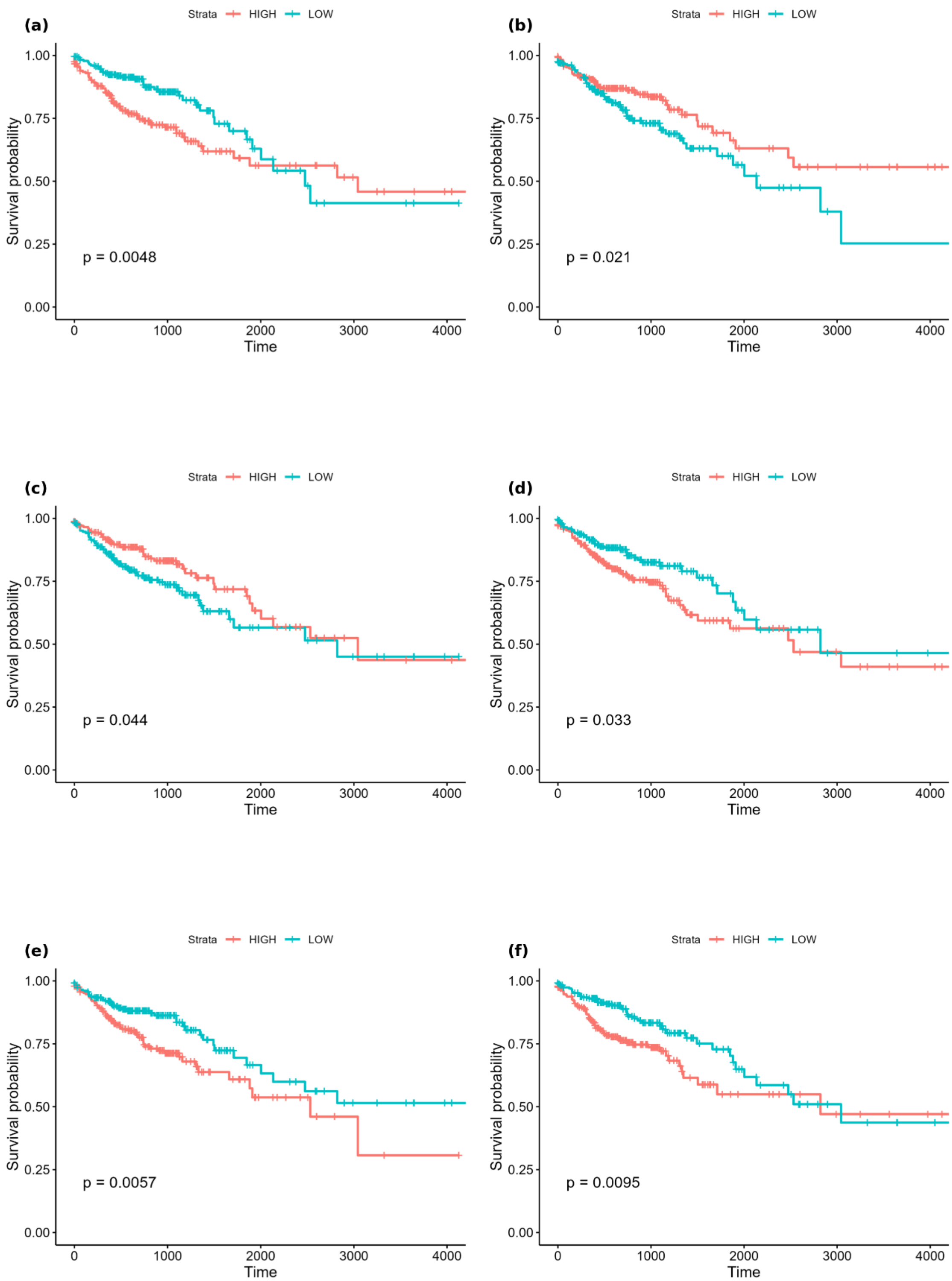

**Supplemental Fig 10.** Kaplan-Meier Plot for BRCA (Gene Expression features) patients stratified by high vs. low expression of (a) *BCL2*, (b) *CXCL1*, (c) *EIF2S2*, (d) *FGD3*, (e) *KDM4B*, (f) *NXNL2*, (g) *SOX11*, (h) *SPARCL1*, and (i) *SUSD3*.



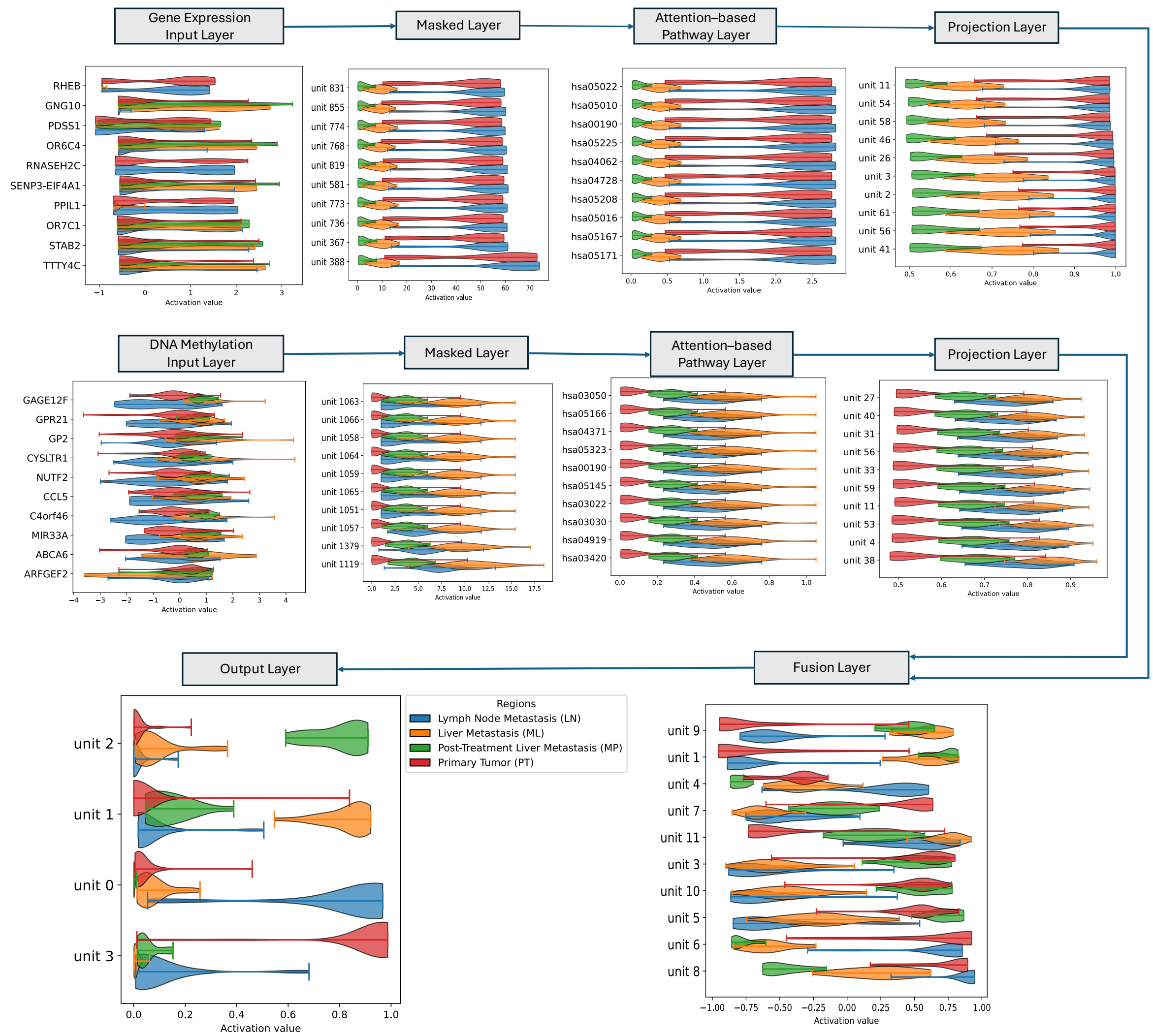

**Supplemental Fig 11.** Kaplan-Meier survival curves for BRCA (DNA Methylation features) patients stratified by high vs. low expression of (a) *IGFALS*, (b) *IGFBP4*, (c) *SCUBE2*, and (d) *SERPINA12*.




**Supplemental Fig 12.** Kaplan-Meier survival curves for COAD (Gene Expression features) patients stratified by high vs. low expression of (a) *ASPHD2* and (b) *COX11*.




**Supplemental Fig 13.** Kaplan-Meier survival curves for COAD (DNA Methylation features) patients stratified by high vs. low expression of (a) *ARRDC1*, (b) *CAMTA1* , (c) *CDC14A* , (d) *DNAJC17* , (e) *FABP4* , (f) *FGF22*




**Supplemental Fig 14.** Activation score of top 10 nodes from each layer of the model trained on scTrio-seq2 data of colorectal cancer. Output layer has 4 nodes.

**References**


1. Ribeiro, M. T., Singh, S. & Guestrin, C. “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining 2016, 1135–1144 (ACM, 2016). https://doi.org/10.1145/2939672.2939778.
2. Cash, A.D. *et al.* Microtubule reduction in Alzheimer’s disease and aging is independent of tau filament formation. *Am. J. Pathol.* 162, 1623–1627 (2003). https://doi.org/10.1016/S0002-9440(10)64296-4
3. Mórotz, G.M. *et al.* Kinesin light chain-1 serine-460 phosphorylation is altered in Alzheimer’s disease and regulates axonal transport and processing of the amyloid precursor protein. *Acta Neuropathol. Commun.* 7, 200 (2019). https://doi.org/10.1186/s40478-019-0857-5
4. Durairajan, S.S.K. *et al.* Unraveling the interplay of kinesin-1, tau, and microtubules in neurodegeneration associated with Alzheimer’s disease. *Front. Cell. Neurosci.* 18, 1432002 (2024). https://doi.org/10.3389/fncel.2024.1432002
5. Mus, E., Hof, P.R. & Tiedge, H. Dendritic BC200 RNA in aging and in Alzheimer’s disease. *Proc. Natl Acad. Sci. USA* 104, 10679–10684 (2007). https://doi.org/10.1073/pnas.0701532104
6. Barranco, N. *et al.* Dense core vesicle markers in CSF and cortical tissues of patients with Alzheimer’s disease. *Acta Neuropathol. Commun.* 9, 140 (2021). https://doi.org/10.1186/s40478-021-01266-9
7. A network pharmacology approach to uncover the key mechanisms of Ginkgo Folium against Alzheimer’s disease. Aging-US (2020).
8. Gene Network Analysis of Alzheimer’s Disease Based on Weighted Gene Co-Expression Network Analysis (WGCNA). PMC (2021).
9. Identifying dysfunctional crosstalk of pathways in various regions of Alzheimer’s disease brains. BioMed Central (2019).
10. Identification of therapeutic targets for Alzheimer’s Disease using machine learning algorithms. Nature (2021).
11. Biomarker Identification for Alzheimer’s Disease Using a Multi-Filter Feature Selection Approach. PMC (2021).
12. Human cytomegalovirus dysregulates neuronal development and function in cerebral organoids. Nature (2022).
13. Potential bioactive compounds and mechanisms of Fibraurea recisa against Alzheimer’s disease: a network pharmacology study. Frontiers (2023).
14. Nixon, R. A. The role of autophagy in neurodegenerative disease. Nat. Med. 19, 983–997 (2013).
15. Tramutola, A., Lanzillotta, C. & Perluigi, M. Targeting mTOR to reduce Alzheimer-related cognitive decline: from current data to novel therapeutic options. Prog. Neurobiol. 184, 101719 (2020).
16. Pei, J. J. & Hugon, J. mTOR-dependent signalling pathways in Alzheimer’s disease. J. Cell. Mol. Med. 12, 2525–2532 (2008).
17. LeBlanc, A. C. The role of apoptotic pathways in Alzheimer’s disease neurodegeneration and cell death. Curr. Alzheimer Res. 2, 389–402 (2005).
18. Heneka, M. T. et al. Innate immune activation in neurodegenerative disease. Nat. Rev. Immunol. 14, 463–477 (2014).
19. Decourt, B., Lahiri, D. K. & Sabbagh, M. N. Targeting tumor necrosis factor alpha for Alzheimer’s disease. Curr. Alzheimer Res. 10, 446–457 (2013).
20. Butterfield, D. A. & Halliwell, B. Oxidative stress, dysfunctional glucose metabolism and Alzheimer disease. Nat. Rev. Neurosci. 20, 148–160 (2019).
21. Kamat, P. K. et al. Glutathione depletion, oxidative stress, and mitochondrial dysfunction in the progression of Alzheimer’s disease. J. Alzheimers Dis. 45, 553–572 (2015).
22. Pimplikar, S. W., Nixon, R. A., Robakis, N. K., Shen, J. & Tsai, L. H. Amyloid-independent mechanisms in Alzheimer’s disease pathogenesis. J. Neurosci. 30, 14946–14954 (2010).

23. Wu, H. Y. et al. Dendritic spine loss and actin cytoskeleton disruption in Alzheimer's disease. Mol. Neurobiol. 55, 7329–7341 (2018).
24. Chang, Y.-C. et al. PPAR-γ agonists reactivate the ALDOC–NR2F1 axis to enhance sensitivity to temozolomide and suppress glioblastoma progression. Cell Commun. Signal. 22, 266 (2024). https://doi.org/10.1186/s12964-024-01645-3
25. Werner, C.K. et al. Expression of the androgen receptor governs radiation resistance in a subset of glioblastomas vulnerable to antiandrogen therapy. Mol. Cancer Ther. 19, 2163–2174 (2020). https://doi.org/10.1158/1535-7163.MCT-20-0095
26. Li, L. et al. Identification of key candidate genes and pathways in glioblastoma by integrated bioinformatical analysis. Exp. Ther. Med. 18, 3439–3449 (2019). https://doi.org/10.3892/etm.2019.7975
27. Zhou, M. et al. Estrogen receptor beta enhances chemotherapy response of GBM cells by down-regulating DNA damage response pathways. Sci. Rep. 9, 6124 (2019). https://doi.org/10.1038/s41598-019-42313-8
28. Del Moral-Morales, A. et al. EZH2 mediates proliferation, migration and invasion promoted by estradiol in human glioblastoma cells. Front. Endocrinol. 13, 703733 (2022). https://doi.org/10.3389/fendo.2022.703733
29. Lino, M.M. & Merlo, A. PI3Kinase signaling in glioblastoma. J. Neuro-Oncol. 103, 417–427 (2011).
30. Lindemann, C. et al. SOCS3 promoter methylation is mutually exclusive to EGFR amplification in gliomas and promotes invasion via STAT3/FAK. Acta Neuropathol. 122, 241–251 (2011).
31. Low, J.T. et al. The lost STING: SLC19A1 downregulation and TMEM173 methylation characterize GBM. Radiother. Oncol. 176, 209–220 (2022).
32. Lin, W. et al. Epigenetic regulation of IL7R shapes immune function and cancer susceptibility. Nature 616, 343–351 (2023).
33. Shukla, S. et al. Methylation silencing of ULK2, an autophagy gene, is essential for astrocyte transformation and tumor growth. J. Biol. Chem. 289, 22306–22318 (2014).
34. May, J.L. et al. IDH3α regulates one-carbon metabolism in glioblastoma. Sci. Adv. 5, eaat0456 (2019).
35. Jung, J. et al. Nicotinamide N-methyltransferase induces cellular hypomethylation and functions as an oncogene in GBM stem cells. JCI Insight 2, e91378 (2017).
36. Jaspers, K. et al. Androgen receptor expression correlates with genome-wide DNA methylation in gliomas and associates with survival. Sci. Rep. 14, 11533 (2024).
37. Tancredi, A. et al. BET inhibition reduces MGMT expression and Pol II/BRD4 binding at the MGMT promoter in GBM. Cell Death Dis. 13, 457 (2022).
38. Bam, M. et al. Genome-wide DNA methylation landscape reveals modulation of T-cell immunity in GBM. BMC Med. Genomics 14, 141 (2021).
39. Ampomah, P.B. et al. Efferocytosis fuels tumor-associated macrophage immunosuppression via methionine-DNMT3A axis. Nat. Metab. 4, 110–125 (2022).
40. Lorimer, I.A.J. et al. Targeting efferocytosis in glioblastoma. Neuro-Oncol. Adv. 6, vdae081 (2024).